Emad Gohari

Senior Machine Learning Engineer · ML Systems, NLP, LLMs

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SUMMARY

Senior machine learning engineer with **8 years of experience** building ML, NLP, and LLM systems from research to production. At Thomson Reuters Labs, shipped **production GenAI summarization and semantic search**, contributed agent configuration and evaluation tooling to a **commercially launched agentic AI product**, and built **model-retraining pipelines and PII-secure data infrastructure** for CLEAR entity resolution. Combines hands-on model development with reproducible experimentation and evaluation, working with researchers, domain experts, and product teams.

TECHNICAL SKILLS

LLMs & Agents: LLM Agents & Tool Use, Agent Harness Engineering, Agentic Frameworks (PydanticAI), Multi-Agent Orchestration, Prompt & Context Engineering, LLM Evaluation (LLM-as-Judge, SME Review), Grounding & Source Citation, Model Selection (Quality, Latency, Cost), GPT-4, Claude, OpenAI API

Machine Learning & NLP: Machine Learning, Deep Learning, NLP, PyTorch, Hugging Face Transformers, Transformer Fine-Tuning (T5), Model Training & Evaluation (F1 Score, ROC-AUC, MRR, MAP, Recall@k), Text Classification & Text Tagging, Clustering Models, Sentence Embeddings, Semantic Search, Information Retrieval, Knowledge Extraction & Representation (Knowledge Graphs, Ontologies), LSTM, scikit-learn, XGBoost, spaCy, NLTK, pandas, NumPy

MLOps, Data & Cloud: MLOps, ML Pipelines, Experiment Tracking (MLflow), Data Pipelines, AWS SageMaker (Studio, Pipelines), Apache Spark (PySpark), Snowflake, AWS (S3, EMR), AWS CloudFormation, PostgreSQL, Elasticsearch, OpenSearch

Languages & Practices: Python, SQL, Java, Scala, Bash, Agile Development

EXPERIENCE

Senior Machine Learning Engineer II (Research) at Thomson Reuters

Jul 2021 - Present

Toronto, Canada

Applied ML at TR Labs for CLEAR, Thomson Reuters' due-diligence platform for KYC/AML screening and investigations, and Checkpoint, its tax research platform. Built LLM features and agent workflows, search models, training pipelines, and secure data infrastructure with researchers, subject-matter experts, and product engineering. Used **SQL/PostgreSQL for entity-resolution data analysis and name/address candidate blocking** before downstream ML matching and ranking.

WORK PROJECTS

- **Entity Resolution Retraining & Snowflake Pipelines: CLEAR** [INTERNAL TOOL] 2026

  Designed and built an **on-demand model-retraining pipeline** on SageMaker Pipelines, turning researchers' feature-generation, training, and evaluation code into reusable DAG steps. Enabled **parallel experiments tracked in MLflow**, cutting experiment turnaround by an **estimated ~40%**. Built proof-of-concept Snowflake processing and training pipelines and assessed the gaps for migrating AWS research workflows to Snowflake.

  Tech & Skills: Python, AWS SageMaker Pipelines, MLflow, PySpark, Snowflake, XGBoost, Poetry

- **Agentic Investigation System: CLEAR Investigate** [PRODUCTION] 2025 - 2026

  Contributed to **CLEAR Investigate, Thomson Reuters' commercially launched agentic AI product** for investigations. Built a **central PydanticAI configuration layer** for system prompts, tool descriptions and availability, and agent/subagent wiring, enabling experimenters to **reconfigure agents from one place** instead of editing scattered definitions. Migrated the team's existing **LLM-as-judge evaluation** capability into an organized repository for reuse and future CI/CD integration.

  Tech & Skills: Python, PydanticAI, Claude, Agent Harness Engineering, Multi-Agent Orchestration, LLM-as-Judge Evaluation, Prompt Engineering

- **CLEAR Business AI: GenAI Report Summarization** [PRODUCTION] 2024

  Built the LLM summarization workflow for CLEAR Business entity reports, **live in production processing hundreds of reports daily**. Designed selective XML extraction for **targeted per-section context** covering business overview, personnel, adverse media, and legal records. Integrated Bing Search and LLM verification to **reduce same-name entity mismatches** and linked claims to report sources. Refined extraction and prompts through SME annotation rounds against exemplars for **10-15 companies**, and compared **GPT-4 and Claude Sonnet on quality, latency, and cost** to inform model selection.

  Tech & Skills: Python, GPT-4, Claude Sonnet, Bing Search API, Prompt & Context Engineering, LLM Evaluation, XML Processing

- **Entity Resolution Data Infrastructure: CLEAR KYC/AML** [INTERNAL TOOL] 2022 - 2024

  Built research data infrastructure for CLEAR's entity-resolution system spanning **~800M entities and billions of documents**, with a **700 idents/second target**. Designed a **PII-isolated AWS environment** with SageMaker Studio, CloudFormation, and per-researcher profiles; defined a **versioned schema reconciling two incompatible annotation sources**, with Spark and Python pipelines to merge records and surface label conflicts; and replaced manual runs with **scheduled, repeatable SageMaker Pipelines**.

  Tech & Skills: Python, Apache Spark (PySpark), AWS SageMaker, AWS CloudFormation

- **Semantic Search Improvement: Checkpoint Tax Research** [PRODUCTION] 2021

  Shipped a **sentence-embedding query-intent classifier** for Checkpoint tax research, distinguishing broad conceptual queries from specific tax lookups to promote higher-level documents. **Improved access to conceptual documents by 95%** within a **~100ms latency budget**. Identified the retrieval gap through SME and customer feedback.

  Tech & Skills: Python, Sentence Embeddings, Text Classification, Semantic Search

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NLP & Machine Learning Engineer at INAGO Inc. Jul 2018 - Jun 2021

Toronto, Canada

Built NLP and language-understanding systems end to end, from data preparation and model training to evaluation and application integration, including transformer fine-tuning and research collaborations with university and industry partners.

WORK PROJECTS

- **Automated Question Generation from Documents** [RESEARCH] 2020

  Fine-tuned the open-weights **T5** model to generate questions from documents, **cutting manual data-curation effort by 40%**. Experimented with input representations and evaluated generated text with BLEURT in a research collaboration with university partners.

  Tech & Skills: Python, PyTorch, Hugging Face Transformers, T5, BLEURT

- **Domain-Specific Language Understanding** [RESEARCH] 2019

  Trained domain-specific **Word2Vec embeddings** and **LSTM-based NLU models**, with interpretability testing to make model decisions transparent.

  Tech & Skills: Python, PyTorch, Word2Vec, LSTM

- **Conversational Question Answering with NLP & LSTM Models** [RESEARCH] 2018

  Developed a **conversational question-answering system** combining syntactic and semantic document analysis with **LSTM-based text tagging**. Used **semi-automatically generated ontologies** to interpret user questions and identify intent, and experimented with knowledge graphs for knowledge representation in an industry research collaboration.

  Tech & Skills: Python, spaCy, LSTM, Text Tagging, Semi-Automatic Ontology Generation, Knowledge Graphs

EDUCATION

M.Sc. in Computer Science, York University 2014 - 2018

Toronto, Canada · GPA: 8.17 / 9

Thesis: Interactive Question Answering Using Frame-based Knowledge Representation

B.Sc. in Computer Engineering, Amirkabir University of Technology 2010 - 2014

Tehran, Iran · GPA: 17.18 / 20

PUBLICATIONS

Question-worthy Sentence Selection for Question Generation

Canadian AI 2020, 2020 · co-authored

Interactive Question Answering Using Frame-based Knowledge Representation

York University M.Sc. Thesis, 2018

Time Aware Topic-based Recommender System

Big Data & Information Analytics, 2016 · co-authored

A Study on Prediction of User's Tendency Toward Purchases in Websites based on Behavior Models

Information and Knowledge Technology (IKT), 6th Conference, IEEE, 2014 · co-authored